Assignment Instructions:

- Use strict mode in all assignments: Add "**use strict**"; at the top of your JavaScript files
- Submit organized code with proper external JS files
- Document your problem-solving approach in comments
- Test all edge cases and document results

Assignment 1: User Greeting Flow

Create a page that implements the following sequence:

1. Show an alert saying "Welcome! Let's get started"
2. Use prompt to ask for the user's email address
3. Use another prompt to ask for their favorite color
4. Display on the page: "Hello [email]! Your favorite color is [color]"
5. Add a confirm dialog asking "Would you like to see this again?"
6. If they click OK, reload the page; if Cancel, show an alert saying "Goodbye!"

Observation Question: What happens to the original page content when you write to it? How can you preserve existing content?

Note: Modify the code to validate that the email contains "@" symbol. If not, keep asking until valid.

Assignment 2: Mathematical Operations with External Functions

Create an external JS file with a function that:

- Takes 3 parameters: two numbers and an operation type ("+", "-", "*", "/")
- Returns the result of the operation (don't use alert/console inside function)
- Create a button that when clicked:
  - Asks user to enter first number
  - Asks user to enter second number
  - Asks user to choose operation (+, -, *, /)
  - Calls your function with these parameters
  - Displays result in the console with format: "Operation: 5 + 3 = 8"

Note: Add input validation inside a loop - keep asking until valid inputs are provided.

Assignment 3: Weather Classification System

Create a function that takes humidity percentage as parameter and prints:

- "Very Dry" if humidity < 30
- "Comfortable" if humidity >= 30 and < 60
- "Humid" if humidity >= 60

Use ternary conditional operator. Can you nest multiple ternary operators? Show your implementation.

Note: Modify to accept both humidity and wind speed, then classify weather as:

- "Pleasant" if humidity < 60 AND wind < 20 km/h
- "Stormy" if both are high
- Use appropriate logic operators (&&, ||)

Assignment 4: Air Quality Index Checker

Create a function that takes 2 parameters: AQI (Air Quality Index) and PM2.5 level:

- Prints "Good Air" if both AQI < 50 AND PM2.5 < 35
- Prints "Moderate" if both are between 50-100 AND 35-75 respectively
- Prints "Unhealthy" if both are above 100 AND 75
- Prints "Data Inconsistent" for any other combination

Critical Thinking Questions:

- Can you use ternary operator here? Why or why not?
- Can you use switch case? Why or why not?
- What's the best control structure for this scenario and why?
- Write the same logic using if-else, then using nested ternary (if possible), compare readability

Note: Add a third parameter for Ozone level and expand the conditions.

Assignment 5: Registration Form Validator

On a registration page, prompt user to enter:

1. **Username**: Must be string, minimum 3 characters, no numbers
2. **Birth Year**: Must be number, between 1950 and 2010
3. **Phone Number**: Must be exactly 11 digits
4. **Email**: Must contain "@" and "."

Requirements:

- For each prompt, validate input
- If invalid, show the same prompt again with error message
- Use loops to keep asking until valid
- After all inputs valid, display on page:

Registration Successful!
Username: john_doe
Birth Year: 1995
Age: 31
Phone: 01234567890
Email: john@example.com

Questions:

1. Which loop type is best for this scenario (for, while, do-while)? Why?
2. How do you check if input is truly a string vs number?
3. What's the difference between typeof and isNaN()?

Assignment 6: Course Eligibility Checker

Create a function that takes student's grade level (parameter) and checks:

- If grade is "A": Show "Eligible for Advanced Programming Course"
- If grade is "B": Show "Eligible for Standard Programming Course"
- If grade is "C": Show "Eligible for Beginner Programming Course"
- If grade is "D": Show "Must take Prerequisite Course first"
- Any other grade: Show "Invalid grade entered"

Requirements:

- Use switch() statement
- Make it case-insensitive (accept both "a" and "A")
- Add a default case

Questions:

1. Why is switch better than if-else here?
2. When would if-else be better than switch?
3. What happens if you forget the break statement?

Challenge: Modify to accept both grade letter AND student's previous course completion status (true/false), adjust eligibility accordingly.

Assignment 7: Safe Expression Evaluator

Create a page with a button that when clicked:

1. Asks user to enter a mathematical expression (Ex: "10+5*2-3")
2. Pass this to a function that:
    - Validates the expression (only numbers and operators +, -, *, /, ())
    - Evaluates it safely
    - Returns result
3. Show alert: "Expression: 10+5*2-3 | Result: 17"

Security Questions:

- Why is using eval() dangerous?
- Research and explain what "code injection" means
- How can you validate the expression before evaluation?
- Implement validation using regex or string checking

Assignment 8: Debugging Practice Lab

Part A: Debug this code and explain all errors:

```
"use strict";
function calculateDiscount(price, discount) {
    if (discount > 100) {
        alert("Invalid discount");
    }
    finalPrice = price - (price * discount / 100);
    return finalPrice;
}

let userPrice = prompt("Enter price:");
let userDiscount = prompt("Enter discount:");
let result = calculateDiscount(userPrice, userDiscount);
console.log("Final Price: $" + result);
```

Part C: Add meaningful console.log() statements:

- Log input types before processing
- Log intermediate calculations
- Log final results
- Compare console debugging vs breakpoint debugging

Questions:

1. List all bugs you found (minimum 4)
2. Classify each as: syntax error, runtime error, or logical error
3. What debugging method was most effective for each type?

Solve day 0, day 1 and day 2 in : Hacker Rank